

c'est quoi l'auto apprentissage et comment il est apparu

Self-learning is a learner-controlled educational process where students construct their own learning programs and perform teaching tasks themselves, which emerged through systematic North American research beginning in the late 1960s and underwent major transformation since the late 20th century due to computing technology, changing work structures, and a cultural shift toward emphasizing individual autonomy in education.

Abstract

Self-learning is defined as "learning in which students construct their own learning program" or "perform teaching tasks themselves", with its core characteristic being learner control over the educational process. Contemporary conceptualizations emphasize autonomy, where "students take control of their learning by defining goals and monitoring progress", and introduce the concept of "apprenance" centered on "mechanisms of appropriation and engagement of the subject". Self-learning involves cognitive processes including motivation, self-regulation, and metacognition, operates through structured phases such as "orientation, planning, testing, monitoring", and incorporates social dimensions with "collective solidarity among peers" even in personal development contexts.

The historical emergence of systematic research on self-directed learning spans "approximately twenty-five years" in North America prior to 1993, with foundational contributions from "A. Tough, M. Knowles, S. Brookfield, H. Long", though its roots extend back to "the time of Alexis de Tocqueville". A major transformation occurred "since the end of the 20th century", marked by a shift from "models centered on intervention and transmission to those focused on appropriation and engagement". This evolution was driven by "challenges posed by the advent of computing, new work structures, and changes in parental and civic roles", with the integration of Information and Communication Technologies making learner autonomy increasingly central. Self-learning has developed across diverse contexts—from formal educational settings to informal associative environments—and applies to learners of all ages, from children in primary schools to adults.

Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "c'est quoi l'auto apprentissage et comment il est apparu"

The search returned 50 total results from Elicit.

We retrieved 50 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Self-Learning Conceptualization:** Does the study define, conceptualize, or theorize self-learning, self-directed learning, autonomous learning, or independent learning?
- **Historical Development:** Does the study examine the historical development or emergence of self-learning concepts, OR does it provide conceptual/theoretical insights about self-learning (if historical development is not the focus)?
- **Study Type Relevance:** Is the study a theoretical paper, conceptual framework, literature review, systematic review, meta-analysis, OR an empirical study that contributes to understanding self-learning concepts?

- **Contextual Scope:** Does the study address self-learning in educational, technological, organizational, or artificial intelligence contexts (rather than being context-free or in irrelevant contexts)?
- **Self-Learning Process Focus:** Does the study address self-learning processes or concepts (rather than focusing solely on specific learning outcomes without discussing self-learning processes)?
- **Substantial Discussion:** Does the study provide substantial discussion of self-learning concepts (rather than only mentioning self-learning incidentally)?
- **Self-Learning Components:** Does the study include self-learning components (rather than focusing exclusively on teacher-directed or instructor-led learning without any self-learning elements)?
- **Conceptual Relevance:** Does the study focus on normal learning processes and conceptual understanding (rather than focusing solely on learning disabilities or pathological learning conditions)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Définition Concept:**

Extract how the study defines or conceptualizes self-learning/auto-apprentissage, including:

- Explicit definitions provided
- Key terms used (autoformation, auto-apprentissage, self-regulation, etc.)
- Essential characteristics or features described
- How it differs from traditional learning approaches
- Scope and boundaries of the concept as presented

- **Développement Historique:**

Extract all information about the historical emergence and evolution of self-learning, including:

- Specific time periods or dates mentioned
- Historical context or social conditions that led to its emergence
- Key historical figures, researchers, or theorists mentioned
- Evolution of the concept over time
- Societal, technological, or educational changes that influenced its development
- Timeline of significant developments or milestones

- **Mécanismes Processus:**

Extract detailed information about how self-learning works and its underlying mechanisms, including:

- Cognitive processes involved (motivation, self-regulation, metacognition)
- Learning strategies and methods described
- Role of autonomy and self-direction
- Psychological or pedagogical mechanisms
- Steps or phases in the self-learning process
- Factors that enable or facilitate self-learning

- **Contextes Milieux:**

Extract information about contexts and environments where self-learning occurs, including:

- Specific settings (associative, educational, technological, professional)
- Target populations or learners described
- Institutional or organizational contexts
- Cultural or social environments
- Age groups or demographic characteristics
- Formal vs informal learning contexts
- **Cadres Théoriques:**

Extract theoretical frameworks, models, or approaches to understanding self-learning, including:

- Specific theories cited or referenced
- Theoretical models or frameworks presented
- Research paradigms or approaches used
- Connections to broader educational or psychological theories
- Conceptual models or diagrams
- Theoretical contributions or innovations proposed

- **Types Formes:**

Extract different types, forms, or variations of self-learning identified, including:

- Specific categories or typologies presented
- Individual vs collective self-learning
- Technology-mediated vs traditional approaches
- Formal vs informal variants
- Domain-specific applications (languages, skills, knowledge areas)
- Developmental or age-related variations

- **Facteurs Influent:**

Extract factors that influence, facilitate, or hinder self-learning, including:

- Individual factors (motivation, prior knowledge, skills)
- Social factors (peer support, cultural context, social environment)
- Institutional factors (policies, resources, support systems)
- Technological factors (tools, platforms, access)
- Barriers or challenges identified
- Success factors or enablers

Results

Characteristics of included studies

Study	Full text retrieved?	Research focus	Study type	Key terminology	Context
P.Robert-Jan Simons et al., 1992	No	Self-regulated learning with computer-aided instruction	Primary study	"auto-apprentissage," "self-regulated learning"	Educational, technological (computer-aided instruction)
B. Dumouchel et al., 2018	No	Self-learning methods and libraries	Review	"auto-apprentissage"	Educational, technological (libraries); adult learners
P. Carré et al., 1993	No	North American research on self-directed learning	Review	"apprentissage auto-dirigé," "autoformation"	Educational; ~25 years of research
P. Carré et al., 2006	No	Self-learning in a culture of "apprenance"	Theoretical analysis	"autoformation," "apprenance"	Institutional contexts; emerged since late 20th century
N. Tremblay et al., 2003	No	Self-learning across contexts	Book synthesis	"autoformation," "autonomie du sujet"	Educational, professional (enterprise)
M. Linard et al., 2003	No	Self-learning, ethics, and technologies	Theoretical analysis	"autoformation," "TIC," "autonomie," "autoregulation"	Initial education, adult training
Pierre-Yves Oudyer et al., 2020	No	Developmental machine learning and autonomous learning	Primary study	"apprentissages autonomes," "curiosité"	Educational (primary schools), technological
P. Portelli et al., 1993	No	Self-learning in associative settings	Primary study	"autoformation," "auto-apprentissage collectif"	Associative (social centers); informal learning
Annie Jézégou et al., 2008	No	Self-directed learning and distance education	Theoretical analysis	"apprentissage autodirigé"	Distance education
D. Ben-Zvi et al., 2007	Yes	Learner autonomy in mathematics and statistics	Primary study	"autonomy," "self-learning," "commognitive perspective"	Educational (school-based); 12-year-old students

The included studies span from 1992 to 2020, with nine studies available only as abstracts and one with full text

retrieved. Research contexts include educational settings (primary, secondary, adult education), technological environments (computer-aided instruction, libraries, robotics), and associative/informal learning contexts. Target populations range from 12-year-old students to adults in various learning environments.

Thematic Analysis

Defining Self-Learning: Conceptualizations and Key Characteristics

Self-learning is conceptualized across multiple dimensions. The most direct definition describes it as "learning in which students construct their own learning program" or "learning in which students perform teaching tasks themselves". This conceptualization emphasizes learner control, with self-directed learning defined as being "based on the degree of learner control over the educational process".

The concept of autonomy emerges as central to understanding self-learning. Learner autonomy is characterized by "students taking control of their learning by defining goals and monitoring progress". This autonomy is sustained when students "treat expert guidance as a starting point for further exploration and rationalization of new concepts", suggesting that autonomy does not mean complete independence from external resources.

Recent conceptualizations introduce the notion of "apprenance," centered on "mechanisms of appropriation and engagement of the subject", which focuses on individual engagement rather than external instruction. This represents a shift from "models centered on intervention and transmission" to those emphasizing personal appropriation of knowledge.

The cognitive and affective dimensions of self-learning are also highlighted. Cognitive learning in self-directed contexts is described as "voluntary" and "often linked to an affective dynamic", with "collective solidarity among peers" playing a crucial role even in personal development. Self-learning involves cognitive processes such as curiosity as "a driving force for autonomous learning" and encompasses "self-organizing learning programs without external rewards or supervision".

Key learning functions associated with self-regulation include "orientation, planning, testing, monitoring", which can be fulfilled through three mechanisms: "action, activation and stimulation". The role of autonomy is characterized by "a high level of autonomy in constructing learning programs", supported by technology such as "computer-aided instruction and linguistic learning programs".

Historical Emergence and Evolution

The historical emergence of self-learning research can be traced through several key periods and developments. In North America, systematic research on self-directed learning spans "approximately twenty-five years" prior to 1993, with foundational contributions from researchers including "A. Tough, M. Knowles, S. Brookfield, H. Long". The concept has deeper roots, with research on associations and their educational role extending "from the time of Alexis de Tocqueville to the present day".

A significant shift occurred "since the end of the 20th century", marked by a "gradual emergence of a new culture of formation influenced by political and sociopedagogical messages" from supranational, European, and national institutions. This period saw a transition from "models centered on intervention and transmission to those focused on appropriation and engagement".

The evolution has been shaped by multiple societal and technological factors. Over "the past thirty years", researchers have studied self-learning "in relation to youth education, corporate training, and popular education", responding to "challenges posed by the advent of computing, new work structures, and changes in parental and civic roles". The

integration of Information and Communication Technologies (TIC) has made autonomy in learning increasingly important, with recent developments focusing on integrating self-learning techniques "into modern deep learning methods and their application in educational settings".

The historical context reveals a shift in educational paradigms. Traditional classrooms have evolved toward "participationist classrooms emphasizing collaboration and autonomy", with increased focus on "communication and individualization of interpersonal communication" and "collaborative learning environments". This evolution reflects broader societal changes in how independence and autonomy are understood and practiced.

Mechanisms and Processes of Self-Learning

Self-learning operates through multiple interconnected mechanisms. At the cognitive level, key processes include "motivation, self-regulation, metacognition", with learners engaging in "scaffolded individualization, exploration of expert discourses". The learning process involves distinct phases: "initial engagement with experts, gradual independence, rationalization of new discourses".

A critical distinction exists between "object-level and meta-level learning", where meta-level learning requires following "experienced individuals to navigate new concepts and rules". This suggests that complete autonomy is not always appropriate; rather, effective self-learning requires "mutual understanding and agreement" and "mutual respect and confidence in expert guidance".

Learning strategies associated with self-directed approaches include "competency-based approaches, problem-based learning, learning contracts, coaching, learning organizations". These methods emphasize "autonomy as a necessary aspect of self-learning" while providing structured support through "diagnostic tools for autonomy".

The technological dimension plays an increasingly important role. Computer-aided instruction functions through "action, activation, and stimulation", with success depending on "teaching students how to use technological tools effectively". Recent developments include "models of autonomous learning motivated by curiosity, use of real robotic platforms" that enable "self-organizing learning programs without external rewards or supervision".

Facilitating factors span multiple domains. Individual factors include "autonomy, flexibility, motivation" and the capacity for "engagement with new discourses". Social factors encompass "peer support, collective solidarity" and "social support and peer solidarity". Institutional support manifests through "structure and support through learning steps like orientation and planning" and "policies promoting autonomy, resources for scaffolding individualization". Technological enablers include "integration of TIC" and tools that facilitate autonomous activity.

Contexts and Applications

Self-learning occurs across diverse settings and populations. Educational contexts range from primary schools to adult education, encompassing both "formal learning environments (education and training)" and informal settings such as "social and socio-cultural centers". Professional contexts include "enterprise" settings where self-learning addresses workplace demands.

Technological environments have become increasingly prominent venues for self-learning. These include "computer-aided instruction", "libraries", and "robotic platforms". Distance education represents a growing context, where self-directed learning theory intersects with transactional perspectives.

Target populations vary significantly by age and circumstance. Studies focus on "12-year-old students" in school settings, while others address "adults" engaging in self-learning across various life circumstances. Research encompasses "children in primary education" as well as learners "accessible to all ages", suggesting self-learning's

applicability throughout the lifespan.

The collective dimension of self-learning emerges particularly in associative contexts, where "micro-milieus educativos (small-scale educational environments)" facilitate learning through "collective solidarity and affective dynamics". These informal contexts demonstrate how self-learning can be simultaneously individual and collective, with "peer solidarity" supporting personal development.

Institutional contexts include "international and national bodies" that shape policy environments, as well as specific settings like "cégeps in Quebec, community colleges in Ontario" and "primary schools". The "Adult Independent Learning Project" exemplifies structured institutional approaches to supporting self-directed learning, with librarians serving as facilitators.

Domain-specific applications include "language skills (computer-aided vocabulary learning program)", "mathematics", and "mathematics and statistics". The breadth of applications demonstrates self-learning's relevance across disciplinary boundaries and learning objectives.

Synthesis

The evidence reveals substantial convergence in how self-learning is conceptualized, despite terminological variations across French ("autoformation," "auto-apprentissage") and English ("self-directed learning," "self-regulated learning") contexts. The core emphasis on learner control and autonomy appears consistently, though with important nuances about the role of external guidance.

A key tension exists between complete learner autonomy and the need for expert scaffolding. This apparent contradiction is resolved by distinguishing between "object-level and meta-level learning". Object-level learning may proceed effectively through autonomous exploration, while meta-level learning—involving new domains or complex concepts—requires "initial engagement with experts, gradual independence". The commognitive perspective suggests that autonomy is not about rejecting authority but about "critically engaging with expert discourse to internalize it as one's own". This reconciles the emphasis on learner autonomy with the recognized need for guidance and social learning processes.

The historical evolution shows a clear progression from isolated early research (de Tocqueville era) through systematic North American studies (approximately 1968-1993) to contemporary integration with technology and institutional policy (late 20th century onward). This trajectory reflects not only theoretical development but also societal changes: computing technology, new work structures, and shifting cultural expectations around learning autonomy. The emergence of self-learning as a major educational approach is thus both a response to practical needs and a reflection of broader values emphasizing individual agency.

Regarding mechanisms, studies converge on the importance of both cognitive processes (motivation, metacognition, curiosity) and social factors (peer support, collective solidarity, mutual understanding). This suggests that effective self-learning is neither purely individualistic nor purely social but involves dynamic interaction between personal agency and social scaffolding. Technology serves as an enabler when properly integrated with pedagogical approaches, but does not replace the need for developing learner competencies and providing appropriate support structures.

The diversity of contexts—from formal schooling to informal associative settings, from childhood to adulthood—demonstrates that self-learning is not confined to specific institutional arrangements or age groups. Rather, it represents a flexible approach adaptable to various circumstances, provided that appropriate conditions are present: learner readiness, social support, institutional resources, and technological tools aligned with pedagogical goals.

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